EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concept | 66 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 1 | Laws of Exponents and Roots

66 questions

Official lessons: 6-1, 6-2, 6-3, 6-4

Formula Bank changes at each Concept boundary.

Concept 1 | Laws of Exponents and Roots OFFICIAL LESSONS 6-1, 6-2, 6-3, 6-4

Laws of Exponents and Roots

Equation-first recall | use the same rail beside every question in this Concept

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q001 | WRITTEN

C06-K01-6-1-WU1

Find: (a) 10^0, (b) 2^{-3}, (c) $(0.2)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V02

Q002 | WRITTEN

C06-K01-6-1-WU2

Let $a \neq 0,\ b \neq 0$. Simplify: (a) $a^{-5}a^{8}$, (b) $a^{5} \div a^{-3}$, (c) $(a^{-1}b)^{2}$, (d) $(2^{2})^{3} \div 2^{-2} \times 2^{-3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q003 | WRITTEN

C06-K01-6-1-TRY1

Find: (a) 5^0, (b) 3^{-2}, (c) $(0.25)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V03

Q004 | WRITTEN C06-K01-6-1-TRY2

Simplify the eight given expressions: a^5a^4, $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, and $2^4 \div (2^{-1})^3 \times 2^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V04

Q005 | WRITTEN

C06-K01-6-1-EX1

Find the nine values: 3^0, 11^0, $(2/3)^0$, 4^{-3}, 5^{-3}, 6^{-2}, $(1/3)^{-3}$, $(3/5)^{-2}$, $(0.04)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V05

Q006 | WRITTEN C06-K01-6-1-EX2

Simplify the twelve same-base expressions in given Question 2: a^2a^4, $a^{-2}a^5$, $a^{-3}a^{-4}$, $a^3 \div a^5$, $a^8 \div a^{-3}$, $a^{-6} \div a^{-2}$, $(a^{-1})^3$, $(a^3b^4)^{-2}$, $(a^{-3}b^2)^{-3}$, $3^5 3^{-3}$, $4^4 4^{-3}$, $5^5 5^{-5}$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V06

Q007 | WRITTEN

C06-K01-6-1-EX3

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^2 3^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^4 2^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, and $(a/2)^3 a^{-4} \div (2a)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V07

Q008 | WRITTEN

C06-K01-6-1-CH

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2}(4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q009 | WRITTEN C06-K01-6-2-WU1

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V02

Q010 | WRITTEN

C06-K01-6-2-WU2

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[6]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q011 | WRITTEN

C06-K01-6-2-TRY1

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V03

Q012 | WRITTEN

C06-K01-6-2-TRY2

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81}-\sqrt[3]{3}+\sqrt[3]{24}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V04

Q013 | WRITTEN | C06-K01-6-2-EX1

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V05

Q014 | WRITTEN

C06-K01-6-2-EX2

Perform the fifteen given calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V06

Q015 | WRITTEN

C06-K01-6-2-CH

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q016 | WRITTEN

C06-K01-6-3-WU1

Express in the form a^r: $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V02

Q017 | WRITTEN

C06-K01-6-3-WU2

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V03

Q018 | WRITTEN C06-K01-6-3-TRY1

Express in the form a^r: $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V04

Q019 | WRITTEN C06-K01-6-3-TRY2

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V05

Q020 | WRITTEN

C06-K01-6-3-EX1

Express the nine given expressions in the form a^r, including roots of a and numerical bases $2,\ 3,\ 5$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V06

Q021 | WRITTEN

C06-K01-6-3-EX2

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/25)^{2/5}]^{-5/4}$, $[(81/100)^{-3/4}]^{2/3}$, $[(64/27)^{-4/3}]^{-1/2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V07

Q022 | WRITTEN

C06-K01-6-3-CH

Simplify $\left[\left((1000/27)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V08

Q023 | WRITTEN C06-K01-6-4-WU

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V09

Q024 | WRITTEN C06-K01-6-4-TRY1

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V10

Q025 | WRITTEN C06-K01-6-4-TRY2

For $a > 0,\ b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V11

Q026 | WRITTEN

C06-K01-6-4-EX1

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V12

Q027 | WRITTEN

C06-K01-6-4-EX2

For $a > 0,\ b > 0$, simplify the six given expressions built from $\sqrt{a}$, $\sqrt[3]{a}$, $\sqrt[4]{a}$, $\sqrt[6]{a}$ and mixed a, b radicals.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Q028 | WRITTEN

C06-K01-6-4-CH

Simplify the ministry expression $\left(\dfrac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}}\right)^{-3/2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F04 | V02

Q029 | WRITTEN

C06-K01-6-4-REF

Assess the statement $\sqrt{a^2} = a$ for all real a, using positive and negative examples.

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q030 | MCQ

C06-K01-6-1-GM01

The value of 2^{-3} is:

A 8

B $1/8$

C -8

D $1/6$

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q031 | MCQ

C06-K01-6-1-GM02

When multiplying equal bases, we:

A subtract exponents

B add exponents

C multiply bases only

D take reciprocals

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

WORKING AREA

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q032 | MCQ C06-K01-6-1-GM03

The expression $a^5 \div a^{-3}$ simplifies to:

A a^2

B a^8

C a^{-8}

D a^{15}

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q033 | MCQ

C06-K01-6-1-GM04

A negative exponent means:

A multiply by -1

B take the reciprocal power

C set the base to zero

D add one

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q034 | MCQ

C06-K01-6-1-GM05

The value of $(0.04)^{-2}$ is:

A 25	**B** 400
C 625	**D** 1600

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control
$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q035 | MCQ

C06-K01-6-1-GM06

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:

A -2

B -1

C 2

D 6

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q036 | MCQ

C06-K01-6-1-GM07

The positive-exponent form of the 6-1 challenge is:

A $a^7 b^6$

B $1/(a^7 b^6)$

C $a^{-7} b^{-6}$

D $1/(a^6 b^7)$

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control
$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q037 | MCQ

C06-K01-6-1-GM08

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:

A $1/3$

B 1

C 3

D 9

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q038 | MCQ C06-K01-6-2-GM01

$\sqrt[3]{64}$ equals:

A 2	**B** 4
C 8	**D** 16

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q039 | MCQ

C06-K01-6-2-GM02

$\sqrt[3]{12}\sqrt[3]{18}$ equals:

A 4	**B** 5
C 6	**D** 9

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q040 | MCQ

C06-K01-6-2-GM03

The fourth root of 1/16 is:

A 1/2

B 1/4

C 2

D 4

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q041 | MCQ

C06-K01-6-2-GM04

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

A $\sqrt[3]{2}$

B $2\sqrt[3]{2}$

C $3\sqrt[3]{2}$

D 0

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q042 | MCQ

C06-K01-6-2-GM05

$\sqrt[4]{625}$ equals:

A 4

B 5

C 5^2

D 25

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q043 | MCQ

C06-K01-6-2-GM06

When combining roots with the same index, multiply:

A the indices

B the radicands

C the exponents only

D the signs only

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q044 | MCQ

C06-K01-6-2-GM07

The roots challenge evaluates to:

A 0	**B** 1
C 2	**D** 4

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q045 | MCQ

C06-K01-6-3-GM01

$\sqrt[5]{2^4}$ in rational-index form is:

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q046 | MCQ

C06-K01-6-3-GM02

The value of $27^{2/3}$ is:

A 3

B 6

C 9

D 18

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q047 | MCQ

C06-K01-6-3-GM03

A reciprocal square root is written as:

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^{2}

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q048 | MCQ

C06-K01-6-3-GM04

The value of $81^{-3/4}$ is:

A 27

B 1/9

C 1/27

D 1/81

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q049 | MCQ

C06-K01-6-3-GM05

The exponent of a in $\sqrt[4]{a^3}$ is:

A $3/4$

B $4/3$

C $-3/4$

D 3

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q050 | MCQ

C06-K01-6-3-GM06

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:

A 4/9

B 9/4

C 16/9

D 27/64

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q051 | MCQ

C06-K01-6-3-GM07

The 6-3 challenge simplifies to:

A 3/10

B 10/3

C 100/9

D 9/100

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q052 | MCQ

C06-K01-6-4-GM01

$2^{2/3}2^{1/2} \div 2^{7/6}$ equals:

A 1/2

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q053 | MCQ

C06-K01-6-4-GM02

$4^{5/6}2^{4/3}$ equals:

A 4

B 6

C 8

D 16

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q054 | MCQ

C06-K01-6-4-GM03

$\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:

A $a^{1/3}$

B $a^{2/3}$

C $a^{5/3}$

D a

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q055 | MCQ C06-K01-6-4-GM04

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:

A 2	**B** 3
C 6	**D** 12

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q056 | MCQ

C06-K01-6-4-GM05

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

A integer powers

B combining rational exponents

C logarithms

D trigonometry

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q057 | MCQ C06-K01-6-4-GM06

The printed decimal in the 6-4 challenge is:

A 0.04

B 0.004

C 0.4

D 4

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Q058 | MCQ

C06-K01-6-4-GM07

For real a, $\sqrt{a^2}$ equals:

A a always

B $-a$ always

C $|a|$

D 0

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Quiz M1 | QUIZ MCQ

C06-K01-Q-M01

Which law gives $a^m a^n = a^{m+n}$?:

A same-base multiplication

B power of a power

C reciprocal law

D root product

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Quiz M2 | QUIZ MCQ

C06-K01-Q-M02

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

A 2

B 5

C 10

D 125

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Quiz M3 | QUIZ MCQ

C06-K01-Q-M03

The value of $64^{-3/2}$ is:

A 512

B 1/64

C 1/512

D 64

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Quiz M4 | QUIZ MCQ

C06-K01-Q-M04

For a negative real a, $\sqrt{a^2}$ is:

A a

B -a

C 0

D a^2

WORKING AREA

FORMULA BANK

Zero / negative
$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws
$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots
$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices
$a^{m/n} = \sqrt[n]{a^m}$

Sign control
$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Quiz W1 | QUIZ WRITTEN

C06-K01-Q-W01

Simplify $a^{-3}a^{7} \div a^{2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Quiz W2 | QUIZ WRITTEN

C06-K01-Q-W02

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Quiz W3 | QUIZ WRITTEN

C06-K01-Q-W03

Evaluate $125^{-2/3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Quiz W4 | QUIZ WRITTEN

C06-K01-Q-W04

Explain why $\sqrt{a^2} = a$ is not true for every real a.

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concept | 50 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 2 | Exponential Functions and Their Applications

50 questions

Official lessons: 6-5, 6-6, 6-7, 6-8, 6-9

Formula Bank changes at each Concept boundary.

Concept 2 | Exponential Functions and Their Applications OFFICIAL LESSONS 6-5, 6-6, 6-7, 6-8, 6-9

Exponential Functions and Their Applications

Equation-first recall | use the same rail beside every question in this Concept

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q001 | WRITTEN

C06-K02-6-5-WU

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1,\ 0,\ 1$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F01 | V02

Q002 | WRITTEN

C06-K02-6-5-TRY

Draw $y = 2^x$ and $y = (1/2)^x$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F01 | V03

Q003 | WRITTEN

C06-K02-6-5-EX

Draw the six given graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ $\quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F01 | V04

Q004 | WRITTEN

C06-K02-6-5-CH

Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q005 | WRITTEN

C06-K02-6-6-WU

Order the given sets 0.7^{-3}, 0.7^{-1}, 0.7^{2} and $5/8$, $5/64$, $6/16$ using inequality signs.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F02 | V02

Q006 | WRITTEN

C06-K02-6-6-TRY

Order the three displayed Try sets: $5^0,\ 5^{-4},\ 5^{3/2}$; $(1/3)^7,\ (1/3)^2,\ (1/3)^{-5}$; and $3^0,\ 3^2,\ 3^{-2}$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V03

Q007 | WRITTEN C06-K02-6-6-EX1

Order the given sets with bases 3, 2, 3/2, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

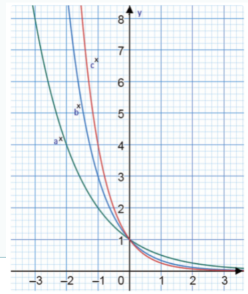
Extrema
$t = a^x \quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V04

Q008 | WRITTEN C06-K02-6-6-CH

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a, b, c.



WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x \quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q009 | WRITTEN

C06-K02-6-7-WU

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V02

Q010 | WRITTEN

C06-K02-6-7-TRY

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V03

Q011 | WRITTEN

C06-K02-6-7-EX

Solve all 24 given equations and inequalities in Exercise (a)–(x), including the $2,\ 3,\ 4,\ 5,\ 8,\ 9,\ 25,\ 27,\ 32,\ 64,\ 81,\ 125$ common-base forms.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V04

Q012 | WRITTEN

C06-K02-6-7-CH

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V05

Q013 | WRITTEN

C06-K02-6-8-WU

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x \quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V06

Q014 | WRITTEN

C06-K02-6-8-TRY

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V07

Q015 | WRITTEN

C06-K02-6-8-EX

Solve the twelve given items (a)–(l), each reducible to a quadratic in 2^x or 3^x.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F03 | V08

Q016 | WRITTEN

C06-K02-6-8-CH

Solve completely for real x: $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \le 0$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q017 | WRITTEN C06-K02-6-9-WU

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x-values.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V02

Q018 | WRITTEN

C06-K02-6-9-TRY

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x-values.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x \quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V03

Q019 | WRITTEN

C06-K02-6-9-EX

Solve the three given extrema models: $-3 \le x \le 2$, $y = 2\cdot 9^{x} - 12\cdot 3^{x} + 7$; $-1 \le x \le 3$, $y = -4 + 2^{2x} - 3^{x+2}$; and the third displayed interval/model.

WORKING AREA

FORMULA BANK

Graph
$y = a^{x} \quad a > 0,\ a \ne 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^{r} < a^{s}$

Common base
$a^{u} = a^{v} \Rightarrow u = v \quad t = a^{x} > 0$

Extrema
$t = a^{x}$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V04

Q020 | WRITTEN

C06-K02-6-9-CH

For $-1 \le x \le 2$, optimise $y = 2 \cdot 4^x - 2^{x+2} + 11$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \ne 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F04 | V05

Q021 | WRITTEN

C06-K02-6-9-REF

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q022 | MCQ

C06-K02-6-5-GM01

Which base gives a strictly decreasing exponential graph?:

A 2

B 3

C 1/2

D π

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ $\quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q023 | MCQ

C06-K02-6-5-GM02

The horizontal asymptote of $y = a^x$ is:

A $x = 0$

B $y = 0$

C $y = 1$

D $x = 1$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q024 | MCQ   C06-K02-6-5-GM03

The range of $y = 5^x$ is:

A $\mathbb{R}$

B $[0, \infty)$

C $(0, \infty)$

D $(-\infty, 0)$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ $\quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q025 | MCQ C06-K02-6-5-GM04

For $y = \pi^x$ on $[-1,\ 2]$, which point is on the graph?:

A $(0, \pi)$

B $(1, \pi)$

C $(2, \pi)$

D $(-1, \pi)$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q026 | MCQ C06-K02-6-6-GM01

For $0 < a < 1$ and $r < s$, which is true?:

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q027 | MCQ

C06-K02-6-6-GM02

The first step when bases differ is to:

A add exponents

B make the bases equal

C take logarithms immediately

D ignore the base

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q028 | MCQ

C06-K02-6-6-GM03

Which is largest? $3^{-2},\ 3^{0},\ 3^{2}$:

A 3^{-2}

B 3^{0}

C 3^{2}

D They are equal

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x \quad t$-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q029 | MCQ C06-K02-6-6-GM04

The challenge term $(\sqrt[3]{4})^3$ equals:

A $2^{3/2}$

B 2^2

C 2^3

D 4^3

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q030 | MCQ C06-K02-6-7-GM01

Solving $4^{x-1} = 8^{x+3}$ gives:

A $x = -11$

B $x = -3$

C $x = 3$

D $x = 11$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q031 | MCQ

C06-K02-6-7-GM02

For $0 < a < 1$, an exponential inequality requires:

A the same exponent order

B a reversed exponent order

C no base conversion

D squaring both sides

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ $\quad$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q032 | MCQ

C06-K02-6-7-GM03

The solution of $3^x = 81$ is:

A 2

B 3

C 4

D 5

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q033 | MCQ

C06-K02-6-7-GM04

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

A $0 \le x \le 2$

B $x > 2$

C $x < -1$

D $x < 2$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q034 | MCQ

C06-K02-6-8-GM05

In substitution methods, $t = a^x$ must satisfy:

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q035 | MCQ — C06-K02-6-8-GM06

The positive root of $t^2 - 3t - 16 = 0$ is:

A $(3-\sqrt{73})/2$

B $(3+\sqrt{73})/2$

C 8

D 16

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q036 | MCQ

C06-K02-6-8-GM07

The challenge quadratic in t is:

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q037 | MCQ C06-K02-6-8-GM08

After solving an exponential quadratic, reject roots with:

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 : \ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q038 | MCQ

C06-K02-6-9-GM01

For extrema, the first required range is the range of:

A x

B $t = a^x$

C a

D y only

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q039 | MCQ

C06-K02-6-9-GM02

The minimum of $y = t^2 - 4t$ at $t = 2$ is:

A -8

B -4

C 0

D 4

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q040 | MCQ

C06-K02-6-9-GM03

The C06-K02 challenge has minimum value:

A 9 at $x = 0$

B 9 at $x = 2$

C 27 at $x = 0$

D 27 at $x = 2$

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q041 | MCQ

C06-K02-6-9-GM04

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:

A 0

B 1

C 2

D It has no minimum

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q042 | MCQ

C06-K02-6-9-GM05

At which value of x does $a^x + a^{-x}$ attain its minimum?:

A $x = -1$

B $x = 0$

C $x = 1$

D It has no stationary point

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Quiz M1 | QUIZ MCQ

C06-K02-Q-M01

Which statement about $y = a^x$ is always true for $a > 0, \quad a \neq 1$?:

A range $(0, \infty)$

B range all reals

C vertical asymptote

D decreasing for every a

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x \quad t$**-range from** x**-domain**

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Quiz M2 | QUIZ MCQ

C06-K02-Q-M02

For base 0.5, increasing the exponent makes the value:

A larger

B smaller

C unchanged

D zero

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1:\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Quiz M3 | QUIZ MCQ

C06-K02-Q-M03

The positive-substitution condition is:

A t<0

B t=0

C t>0

D t can be any real

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Quiz M4 | QUIZ MCQ C06-K02-Q-M04

For $y = 2(t-1)^2 + 9$, the minimum is:

A 0	**B** 1
C 9	**D** 11

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ **t-range from x-domain**

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Quiz W1 | QUIZ WRITTEN

C06-K02-Q-W01

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Quiz W2 | QUIZ WRITTEN

C06-K02-Q-W02

Order $2^{-3},\ 2^{0},\ 2^{5/2}$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Quiz W3 | QUIZ WRITTEN

C06-K02-Q-W03

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ $\quad$ t-range from x-domain

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Quiz W4 | QUIZ WRITTEN C06-K02-Q-W04

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

WORKING AREA

FORMULA BANK

Graph
$y = a^x \quad a > 0,\ a \neq 1$

Growth / decay
$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$

Magnitude
$a > 1 :\ r < s \Rightarrow a^r < a^s$

Common base
$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$

Extrema
$t = a^x$ t-range from x-domain

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MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concept | 74 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 3 | Introduction to Logarithms and Their Properties

74 questions

Official lessons: 6-10, 6-11, 6-12, 6-13, 6-14, 6-15, 6-16, 6-17

Formula Bank changes at each Concept boundary.

Concept 3 | Introduction to Logarithms and Their Properties
OFFICIAL LESSONS 6-10, 6-11, 6-12, 6-13, 6-14, 6-15, 6-16, 6-17

Introduction to Logarithms and Their Properties

Equation-first recall | use the same rail beside every question in this Concept

Definition

$$a^p = M \iff \log_a M = p$$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q001 | WRITTEN

C06-K03-6-10-WU

Convert the given forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V02

Q002 | WRITTEN

C06-K03-6-10-TRY

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V03

Q003 | WRITTEN

C06-K03-6-10-EX

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V04

Q004 | WRITTEN

C06-K03-6-10-CH

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V05

Q005 | WRITTEN

C06-K03-6-11-WU

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1,\ \log_5 5,\ \log_2 8,\ \log_2 16,\ \log_5(1/5),\ \log_7(49^{3/2})$.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \Longleftrightarrow \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V06

Q006 | WRITTEN

C06-K03-6-11-TRY

Find the six Try values $\log_5 1,\ \log_3 3,\ \log_4 4,\ \log_3 27,\ \log_2(1/4),\ \log_5(1/125)$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V07

Q007 | WRITTEN

C06-K03-6-11-EX

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V08

Q008 | WRITTEN

C06-K03-6-11-CH

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q009 | WRITTEN C06-K03-6-12-WU

Perform $\log_6 2 + \log_6 3$ and $\log_2 30 - \log_2 3 - \log_2 5$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V02

Q010 | WRITTEN

C06-K03-6-12-TRY

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V03

Q011 | WRITTEN

C06-K03-6-12-EX

Perform all 18 given expressions (a)+(r), combining same-base logarithms exactly.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V04

Q012 | WRITTEN

C06-K03-6-12-CH

Simplify the fractional expression $\log_5 75 + 3\log_5 \sqrt{2} - 2\log_5 \sqrt{6}$ to one reduced fraction.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V05

Q013 | WRITTEN C06-K03-6-15-WU

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24,\ \log_{10}(27/4),\ \log_{10} 45,\ \log_{10}(24/75)$ in terms of $a,\ b$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V06

Q014 | WRITTEN C06-K03-6-15-TRY

Given $\log_{10} 2 = a,\ \log_{10} 3 = b$, express the four Try arguments $72,\ 3^8/10,\ 15,\ 12$ in terms of $a,\ b$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V07

Q015 | WRITTEN C06-K03-6-15-EX

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V08

Q016 | WRITTEN

C06-K03-6-15-CH

If $\log_c 3 = x,\ \log_c 2 = y$, write $\log_c 12,\ \log_c(3/8),\ \log_c \sqrt{6},\ \log_2 3,\ \log_{3^b}(2^b c)$ in terms of $x,\ y$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q017 | WRITTEN C06-K03-6-13-WU

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V02

Q018 | WRITTEN C06-K03-6-13-TRY

Evaluate the five Try logarithms using a suitable common base.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V03

Q019 | WRITTEN

C06-K03-6-13-EX

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V04

Q020 | WRITTEN

C06-K03-6-13-CH

Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V05

Q021 | WRITTEN

C06-K03-6-14-WU

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V06

Q022 | WRITTEN C06-K03-6-14-TRY

Solve the four Try products of logarithms after making the bases the same.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V07

Q023 | WRITTEN

C06-K03-6-14-EX

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V08

Q024 | WRITTEN

C06-K03-6-14-CH

Simplify $\log_x(\sqrt{y})\ \ \log_{y^2}(z^3)\ \ \log_{\sqrt{z}}(x^4)$ for $x > 1$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V09

Q025 | WRITTEN

C06-K03-6-14-REF

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q026 | WRITTEN

C06-K03-6-16-WU

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V02

Q027 | WRITTEN

C06-K03-6-16-TRY

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and 1/3.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V03

Q028 | WRITTEN

C06-K03-6-16-EX

Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V04

Q029 | WRITTEN C06-K03-6-16-CH

For $f(x) = \log_5(x-2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V05

Q030 | WRITTEN C06-K03-6-17-WU

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2\log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V06

Q031 | WRITTEN C06-K03-6-17-TRY

Order the four Try sets of logarithms using monotonicity and change of base.

WORKING AREA

FORMULA BANK

Definition
$$a^p = M \iff \log_a M = p$$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V07

Q032 | WRITTEN

C06-K03-6-17-EX

Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V08

Q033 | WRITTEN

C06-K03-6-17-CH

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q034 | MCQ C06-K03-6-10-GM01

$\log_2 32$ equals:

A 2

B 4

C 5

D 32

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q035 | MCQ C06-K03-6-10-GM02

$\log_{1/2} 2$ equals:

A -2	**B** -1
C 1	**D** 2

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q036 | MCQ

C06-K03-6-11-GM03

$\log_5(1/125)$ equals:

A -3	**B** -2
C 2	**D** 3

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q037 | MCQ C06-K03-6-11-GM04

$\log_7 49^{3/2}$ equals:

A 1	**B** 2
C 3/2	**D** 7/2

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q038 | MCQ

C06-K03-6-12-GM01

$\log_a M + \log_a N$ equals:

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q039 | MCQ

C06-K03-6-12-GM02

The coefficient $3\log_5 2$ becomes:

A $\log_5(2+3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q040 | MCQ

C06-K03-6-15-GM03

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:

A $a + b$

B $2a + 3b$

C $3a + b$

D $3a + 2b$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q041 | MCQ

C06-K03-6-15-GM04

$\log_c \sqrt{6}$ in terms of $x = \log_c 3,\ y = \log_c 2$ is:

A $x + y$

B $(x + y)/2$

C $2x + y$

D $x + 2y$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q042 | MCQ C06-K03-6-13-GM01

$\log_{16} 64$ equals:

A 1/2	**B** 3/2
C 2	**D** 4

WORKING AREA

FORMULA BANK

Definition
$$a^p = M \iff \log_a M = p$$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q043 | MCQ

C06-K03-6-13-GM02

Change of base gives $\log_a b =$:

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q044 | MCQ

C06-K03-6-14-GM03

$\log_4 16 \cdot \log_9 3$ equals:

A 1/2

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Definition

$a^p = M \Longleftrightarrow \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws

$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q045 | MCQ

C06-K03-6-14-GM04

The challenge product of three logarithms equals:

A 6	**B** 8
C 12	**D** 24

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q046 | MCQ

C06-K03-6-16-GM01

The domain of $y = \log_a x$ is:

A all real x

B $x \geq 0$

C $x > 0$

D $x \neq 0$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q047 | MCQ

C06-K03-6-16-GM02

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q048 | MCQ C06-K03-6-17-GM03

For $a > 1$ and $M < N$, which is true?:

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Cannot compare

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q049 | MCQ

C06-K03-6-17-GM04

For $0 < a < 1$ and $M < N$, which is true?:

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Both are positive

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q050 | MCQ

C06-K03-6-16-GM05

The vertical asymptote of $y = \log_a x$ is:

A $x = 0$

B $y = 0$

C $x = 1$

D $y = 1$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q051 | MCQ

C06-K03-6-17-GM06

A negative coefficient $-2\log_a M$ can be written as:

A $\log_a M^{-2}$

B $\log_a(-2M)$

C $\log_a(M-2)$

D $2\log_a M$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q052 | MCQ

C06-K03-6-10-GM05

The base of $\log_a M$ is:

A M

B a

C p

D $1/a$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q053 | MCQ — C06-K03-6-10-GM06

The argument of $\log_a M$ is:

A a

B M

C p

D 0

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q054 | MCQ

C06-K03-6-11-GM07

$\log_2(1/8)$ equals:

A -3	**B** -2
C 2	**D** 3

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q055 | MCQ C06-K03-6-11-GM08

$\log_4 4$ equals:

A 0

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q056 | MCQ

C06-K03-6-11-GM09

$\log_3 1$ equals:

A -1

B 0

C 1

D 3

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q057 | MCQ

C06-K03-6-12-GM05

$\log_a M - \log_a N$ equals:

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M - N)$

D $\log_a(N/M)$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q058 | MCQ

C06-K03-6-12-GM06

$2\log_3 5$ equals:

A $\log_3 10$

B $\log_3 25$

C $\log_3 7$

D $\log_3(5/2)$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q059 | MCQ

C06-K03-6-15-GM07

If $a = \log_{10} 2,\ b = \log_{10} 3, \log_{10} 12$ is:

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties F02 | V01

Q060 | MCQ C06-K03-6-15-GM08

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

A a

B $1 - a$

C $1 + a$

D $-a$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q061 | MCQ C06-K03-6-13-GM05

$\log_8 32$ equals:

A 2/3	**B** 3/2
C 5/3	**D** 3

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q062 | MCQ C06-K03-6-13-GM06

$\log_{25} 125$ equals:

A 1/2

B 3/2

C 2

D 5/2

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q063 | MCQ

C06-K03-6-14-GM05

$\log_a b \cdot \log_b a$ equals:

A 0

B 1

C 2

D $\log_a b$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q064 | MCQ

C06-K03-6-14-GM06

$\log_2 8 \cdot \log_8 2$ equals:

A 1/3

B 1

C 3

D 8

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q065 | MCQ

C06-K03-6-14-GM07

The best common base for 16 and 64 is:

A 2

B 4

C 8

D 10

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q066 | MCQ C06-K03-6-16-GM07

The range of $y = \log_a x$ is:

A $x > 0$

B $\mathbb{R}$

C $[0, \infty)$

D $y > 0$

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Quiz M1 | QUIZ MCQ

C06-K03-Q-M01

The logarithmic equivalent of $a^p = M$ is:

A $\log_a M = p$

B $\log_p M = a$

C $\log_M a = p$

D $\log_a p = M$

WORKING AREA

FORMULA BANK

Definition

$a^p = M \iff \log_a M = p$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Quiz M2 | QUIZ MCQ

C06-K03-Q-M02

$\log_2 8 + \log_2 4$ equals:

A 2

B 3

C 5

D 12

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \Longleftrightarrow \log_a M = p$$

Values

$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws

$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Quiz M3 | QUIZ MCQ | C06-K03-Q-M03

$\log_{16} 64$ equals:

A 1/2

B 3/2

C 2

D 4

WORKING AREA

FORMULA BANK

Definition
$a^p = M \Longleftrightarrow \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Quiz M4 | QUIZ MCQ

C06-K03-Q-M04

For $a > 1$, $M < N$ implies:

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Quiz W1 | QUIZ WRITTEN

C06-K03-Q-W01

Find $\log_3(1/27)$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Quiz W2 | QUIZ WRITTEN C06-K03-Q-W02

Simplify $\log_5 25 + \log_5 5$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Quiz W3 | QUIZ WRITTEN C06-K03-Q-W03

Find $\log_{16} 64$ using change of base.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Quiz W4 | QUIZ WRITTEN

C06-K03-Q-W04

State the domain and range of $y = \log_a x$.

WORKING AREA

FORMULA BANK

Definition
$a^p = M \iff \log_a M = p$

Values
$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$

Laws
$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$

Change base
$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs
$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$

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MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concepts | 48 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed
Assistant Lecturer • Faculty of Engineering • Cairo University
01120009622

Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 4 | Applications of Logarithms

48 questions

Official lessons: 6-18, 6-19, 6-20, 6-21, 6-22

The Formula Bank changes at each Concept boundary.

Concept 4 | Applications of Logarithms **OFFICIAL LESSONS 6-18, 6-19, 6-20, 6-21, 6-22**

Applications of Logarithms

Equation-first recall | use the same rail beside every question in this Concept

Log equations

$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities

$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution

$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema

$t = \log_a x \quad t$-range from the x-domain

Common log

$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Applications of Logarithms

F01 | V01

Q001 | WRITTEN

C06-K04-6-18-WU

Solve $\log_2 x = 4$.

WORKING AREA

FORMULA BANK

Log equations

$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities

$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution

$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema

$t = \log_a x$ **t-range from the x-domain**

Common log

$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F01 | V02

Q002 | WRITTEN

C06-K04-6-18-TRY

Solve $\log_2(2x-1)=\log_2 3$.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F01 | V03

Q003 | WRITTEN

C06-K04-6-18-EX

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1:\ M < N \qquad 0 < a < 1:\ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F01 | V04

Q004 | WRITTEN C06-K04-6-18-CH

Solve $\log_3(x+2) - \log_9((x-2)^2) = 1$ completely.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F02 | V01

Q005 | WRITTEN C06-K04-6-19-WU

Solve $\log_3 x < \log_3 81$ and $\log_{1/2}(x+2) + \log_{1/2}(x-5) > -3$.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F02 | V02

Q006 | WRITTEN

C06-K04-6-19-TRY

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F02 | V03

Q007 | WRITTEN

C06-K04-6-19-EX

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F02 | V04

Q008 | WRITTEN

C06-K04-6-19-CH

Solve the given challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \; M < N \qquad 0 < a < 1 : \; M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F03 | V01

Q009 | WRITTEN

C06-K04-6-20-WU

Solve $5(\log_2 x)^2 - 16\log_2 x + 3 < 0$.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1:\ M < N \qquad 0 < a < 1:\ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F03 | V02

Q010 | WRITTEN

C06-K04-6-20-TRY

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F03 | V03

Q011 | WRITTEN

C06-K04-6-20-EX

Solve the given groups: three equations, three inequalities, and three exact logarithmic values.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \; M < N \qquad 0 < a < 1: \; M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F03 | V04

Q012 | WRITTEN — C06-K04-6-20-CH

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ for the positive wave-intensity multiplier x.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | V01

Q013 | WRITTEN

C06-K04-6-21-WU

For $y = -(\log_3 x)^2 + 4\log_3 x + 3$, $1 \le x \le 27$, find maximum/minimum values and the corresponding x.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | V02

Q014 | WRITTEN

C06-K04-6-21-TRY

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 :\ M < N \qquad 0 < a < 1 :\ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F04 | V03

Q015 | WRITTEN C06-K04-6-21-EX

Solve the three given extrema problems, each with a different base and interval.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1:\ M < N \qquad 0 < a < 1:\ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F04 | V04

Q016 | WRITTEN C06-K04-6-21-CH

For $y = -(\log_2 x)^2 + 2\log_2 x + 4,\ 1 \le x \le 8$, find the maximum and minimum health ratings and their concentrations.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x \quad t$-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | V05

Q017 | WRITTEN

C06-K04-6-22-WU

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F04 | V06

Q018 | WRITTEN C06-K04-6-22-EX

Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | V07

Q019 | WRITTEN C06-K04-6-22-CH

Using the printed common logs, determine the number of digits and the leading digit of 12^{50}.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | V08

Q020 | WRITTEN

C06-K04-6-22-REF

Solve $\log_x(x+2) \leq \log_x(4-x)$ for all real x.

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x \quad$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q021 | MCQ

C06-K04-6-18-GM01

The argument condition for $\log_3(x-5)$ is:

A x<5

B $x \geq 5$

C x>5

D all real x

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q022 | MCQ

C06-K04-6-18-GM02

If $\log_2(2x-1)=\log_2 3$, then $x=$:

A 1

B 2

C 3

D 4

WORKING AREA

FORMULA BANK

Log equations
$\log_a M=\log_a N \Rightarrow M=N \quad (M,N>0)$

Log inequalities
$a>1:\ M<N \qquad 0<a<1:\ M>N$

Substitution
$t=\log_a x \quad t\in\mathbb{R} \quad x=a^t$

Extrema
$t=\log_a x$ **t-range from the x-domain**

Common log
$n-1<\log_{10}N<n\Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q023 | MCQ

C06-K04-6-18-GM03

The valid solution of the challenge equation is:

A 1	**B** 2
C 4	**D** 8

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q024 | MCQ C06-K04-6-18-GM04

For same-base logarithms, equality of logs gives:

A arguments equal

B bases equal

C arguments add

D no result

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \; M < N \qquad 0 < a < 1: \; M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q025 | MCQ

C06-K04-6-19-GM05

For base $a > 1$, $\log_a M < \log_a N$ means:

A M>N

B M<N

C M=N

D MN<0

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1:\ M < N \qquad 0 < a < 1:\ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q026 | MCQ

C06-K04-6-19-GM06

For base $0 < a < 1$, the inequality direction is:

A preserved

B reversed

C removed

D squared

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q027 | MCQ

C06-K04-6-19-GM07

The domain of $\log_{1/2}(x-5)$ requires:

A x<5

B $x \leq 5$

C x>5

D $x \neq 5$

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q028 | MCQ

C06-K04-6-19-GM08

The warm-up intersection is:

A 0<x<5

B 5<x<6

C x>6

D -3<x<6

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q029 | MCQ   C06-K04-6-20-GM09

The best substitution for a quadratic logarithmic equation is:

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D t=1/x

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \; M < N \qquad 0 < a < 1 : \; M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q030 | MCQ

C06-K04-6-20-GM10

The inequality $5t^2 - 16t + 3 < 0$ gives:

A $t<1/5$	**B** $1/5<t<3$
C $t>3$	**D** $0<t<1$

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q031 | MCQ

C06-K04-6-20-GM11

The substitution variable must satisfy:

A t>0

B $t \in \mathbb{R}$

C t<0

D t=1

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q032 | MCQ

C06-K04-6-20-GM12

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

A x<1/9

B 1/9<x<27

C x>27

D 0<x<3

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x \quad$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q033 | MCQ

C06-K04-6-21-GM13

For $y = -(t-2)^2 + 7$, the maximum occurs at:

A t=0

B t=1

C t=2

D t=3

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q034 | MCQ

C06-K04-6-21-GM14

The correct first step for extrema is:

A differentiate in x

B $set t = \log_a x$

C ignore the interval

D take x=0

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q035 | MCQ

C06-K04-6-21-GM15

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:

A [-1,3]

B [0,3]

C [1,27]

D all real

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1 : \ M < N \qquad 0 < a < 1 : \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x \quad$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q036 | MCQ

C06-K04-6-21-GM16

The water-tank challenge maximum health rating is:

A 3

B 4

C 5

D 8

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \; M < N \qquad 0 < a < 1: \; M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ **t-range from the x-domain**

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ **digits**

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q037 | MCQ

C06-K04-6-22-GM17

The number of digits of N is determined by:

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C N+1

D 10^N

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ t-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q038 | MCQ

C06-K04-6-22-GM18

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:

A 8th

B 9th

C 10th

D 11th

WORKING AREA

FORMULA BANK

Log equations
$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

Log inequalities
$a > 1: \ M < N \qquad 0 < a < 1: \ M > N$

Substitution
$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$

Extrema
$t = \log_a x$ $\quad t$-range from the x-domain

Common log
$n - 1 < \log_{10} N < n \Rightarrow n$ digits

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q039 | MCQ

C06-K04-6-22-GM19

The common logarithm base is:

A 2

B e

C 10

D any base

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Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q040 | MCQ

C06-K04-6-22-GM20

The solution set of $\log_x(x+2) \leq \log_x(4-x)$ is:

A (0,1)

B (1,4)

C (0,4)

D empty

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Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VQUIZ

Quiz M1 | QUIZ MCQ C06-K04-Q-M01

Solve $\log_5(x-1)=2$:

A x=26

B 0

C 1

D cannot be determined

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Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C06-K04-Q-M02

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$:

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

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Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VQUIZ

Quiz M3 | QUIZ MCQ

C06-K04-Q-M03

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$:

A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

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Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VQUIZ

Quiz M4 | QUIZ MCQ

C06-K04-Q-M04

Find the maximum of $y = -(\log_3 x)^2 + 2\log_3 x + 4$ on $1 \le x \le 27$:

A 5 at x=3

B 0

C 1

D cannot be determined

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Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C06-K04-Q-W01

Solve $\log_3(x+1) = 2$.

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Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VQUIZ

Quiz W2 | QUIZ WRITTEN

C06-K04-Q-W02

Solve $\log_2 x \geq 3$.

WORKING AREA

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Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VQUIZ

Quiz W3 | QUIZ WRITTEN

C06-K04-Q-W03

Solve $(\log_5 x)^2 - 4 < 0$.

WORKING AREA

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Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VQUIZ

Quiz W4 | QUIZ WRITTEN C06-K04-Q-W04

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010,\ \log_{10} 3 = 0.4771$?

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